The thermo-acoustic response of a premixed swirl burner

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Abstract: Application of the theory of thermo-acoustic excitation of flames requires a detailed knowledge of the flame response to plane-wave velocity fluctuations of a given frequency through the burner. A model of the type of swirl burner used in some designs of industrial gas turbine is investigated, chemiluminescent emission being used as a marker of the heat release from both methane and propane premixed flames. The effects of the equivalence ratio and the withdrawal of the swirler so as to allow an annulus of unswirled premixture are also examined.

It is found that there are two distinct modes of self-excitation. The response of different burner configurations in the mode with the higher frequency is analysed in terms of four parameters. These characterize the position and spread of the axial distribution of heat release fluctuations, and the origin and effective velocity of convection of disturbances.

From these parameters, the Rayleigh integral can be calculated and this defines the frequency range over which excitation is possible. If the acoustic conditions require that the flame is lengthened, retracting the swirler, reducing the equivalence ratio or changing from propane to methane are all shown to be possible control measures for this particular burner.

Keywords: premixed flame, thermo-acoustics, Rayleigh criterion, swirl burner

NOTATION

A	flow area
c	speed of sound
Co	coherence of two signals, e.g. $Co(p, q) = [\lvert\overline{pq}\rvert/(\tilde{p}\ \tilde{q})]^2$
d	diameter of the burner outlet
f	frequency
K	mean pressure loss coefficient
M	Mach number
p	fluctuating acoustic pressure
P	mean pressure
q	fluctuating heat release per unit volume
Q_a	volumetric flowrate of air
s	standard deviation in a Gaussian distribution for fluctuating heat release
Tr	transfer function between two signals, e.g. $Tr(q/p) = \overline{pq}/p^2$
u	fluctuating acoustic velocity
u_1	fluctuating acoustic velocity at the burner outlet
U	mean axial velocity through the burner outlet
U_{eff}	effective axial convection velocity of heat release fluctuations
x	axial distance from burner outlet
x_0	effective origin of heat release fluctuations
x_1	centre of the Gaussian distribution for fluctuating heat release
Z_1	acoustic impedance of the supply system at burner outlet $\equiv p_f/(\rho u_1 c)$
γ	ratio of specific heat capacities for combustion products
ρ	mean density of gas
ϕ	equivalence ratio
$\phi_{u/p}$ or $Ph(u/p)$	equivalence ratio phase of u relative to p, i.e. phase of $Tr(u/p)$

Subscripts

f	of the combustion chamber at the microphone position
w	of the air supply system at the microphone position

Superscripts

-	time-averaged
~	r.m.s. value

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1 INTRODUCTION

The current practice of firing lean premixtures of natural gas in gas turbines has led to a need for greater understanding of the problem of combustion-induced vibrations. This is due to the susceptibility of such systems (which are operated in the lean mode to suppress NO_x formation) to thermo-acoustic interactions, and the possibility that major structural damage will result. The susceptibility arises from the acoustic characteristics of the premixture supply system and the combustion chamber, and from the sensitivity of the flame to the velocity fluctuations associated with plane acoustic waves.

The principles of excitation are well known, having been enunciated by Lord Rayleigh. His criterion for excitation at a particular frequency may be refined to the form

$$\frac{\gamma - 1}{\gamma \overline{P}} \int_0^{\infty} \int |p||q| \cos \ \phi_{q/p} \ \mathrm{d}x \ \mathrm{d}A \geqslant \text{acoustic losses}$$

where $\overline{P}$ is the mean pressure, p is the acoustic pressure fluctuation, which can be shown to be essentially constant in the main flame zone in the long wavelength limit, q is the fluctuating component of heat release per unit volume, $\phi_{q/p}$ is the phase angle between them, and A is the cross-sectional area of the flame at axial distance x. If the criterion is satisfied for a particular acoustic mode, the flame will 'self-excite' and the magnitude of the fluctuations will build up until non-linear processes in the flame limit the heat release fluctuation and the acoustic losses become such that the equality is satisfied. At frequencies close to (but not identical with) the resonant frequencies of the combustion chamber, little energy input is required for very high levels of pressure fluctuation to be attained.

While the principles are established, application of the theory to a practical system demands detailed knowledge of the flame response. A number of contributions in the literature have presented semi-empirical models of the response of simple flames, and they have been reviewed by Raun *et al.* [**1**]. An early noteworthy example was the work of Merk [**2**] on a Bunsen-type flame, since he introduced a convective time-delay concept. More recently, Keller and Saito [**3**] and Buechner *et al.* [**4**] have correlated data from dump combustion systems. Shih *et al.* [**5**] presented detailed measurements of the Rayleigh integral for such a system but did not attempt to correlate them. Macquisten and Dowling [**6**] postulated a model for a V-shaped flame and used it in conjunction with the acoustic characteristics of their system to produce a complete solution, which compared favourably with experimental results.

This must be the goal for more complex systems, and the mathematical framework has been set out in papers by Lieuwen and Zinn [**7**], Hobson *et al.* [**8**], Abbott [**9**] and Dowling and Hubbard [**10**]. Sivasegaram and Whitelaw [**11**] have noted the complex effects that swirl can have on the regimes of excitation. Thus, realistic flame models are required for swirl burners. Although computational fluid dynamics (CFD) may make a contribution to this (see, for example, reference [**12**]), comprehensive data on the flame response, such as those recently presented by Broda *et al.* [**13**] and Paschereit *et al.* [**14**], are needed. The present contribution is directed to this end, with the correlation of a large number of sets of measurements on a small model. The model was of the type of swirl burner found in some commercial combustion systems.

Most of the data in the investigations already mentioned were obtained by recording chemiluminescent emission as a marker of the heat release. This principle was also used here, building on the advanced techniques and analysis which have previously been deployed on a bluff-body stabilized flame [**15**, **16**]. These are briefly described and then the analysis of the acoustics is outlined, before the presentation of the results for the various flames. Finally, these are discussed in the context of a semi-empirical model which allows calculation of the Rayleigh integral.

The data are unique in that the advanced ultraviolet (UV) techniques have been used to demonstrate the structure (both axial and radial) of the response, at specific frequencies, of flames that are close in configuration to those of practical relevance. From this knowledge base, the effect of changes in fuel, equivalence ratio and burner geometry on the acoustic response of the flame have been investigated and related to the acoustic properties of the premixture supply system. Such a comprehensive approach to the thermo-acoustics of a swirl burner has not been published previously.

2 APPARATUS

A diagram of the burner, air and fuel supply system, and combustion chamber is shown in Fig. 1. The air was supplied from the laboratory compressed air system through a constriction to the side of a pipe of 51 mm diameter with the end blanked off. This effectively ensured an acoustic velocity node at the end of the pipe. The air then flowed round a bend to the burner mouth through a leaf valve and a tapered section in which a 16-vane swirler was located. The swirler operated in one of four positions for different experiments:

Configuration 1. This is the original configuration, not fully seated on the tapered section.

Configuration 2. The swirler in this configuration was fully seated.

Configuration 3. In this configuration the swirler is

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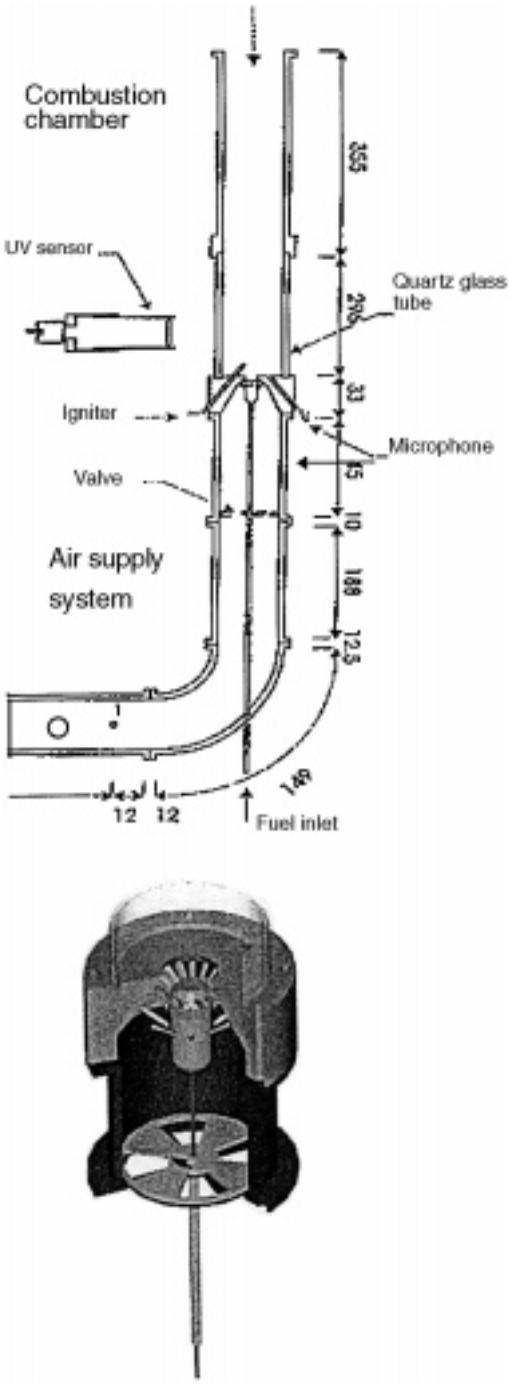


Fig. 1 The premixed swirl burner

retracted from the taper by 0.75 mm, so that some air passed through the annulus outside the swirler.

Configuration 4. Finally the swirler is further retracted from the tapered section, by 1.5 mm.

When the swirler was properly seated, the swirl number was calculated to be about 0.4.

The fuel was piped to a small plenum below the swirler from which it was injected into the air flow through a tree of eight pipes, each branch of which had five downstream-facing holes in it. The fuel was thus entrained into the air in the wake of the branch before the mixture was accelerated through the swirler. By the time that the air and fuel left the burner through an orifice of 19 mm diameter, full premixing was assumed, there being no fine structure in the resulting flame. In addition, there was a 1.2 mm hole in the top of the plenum through which undiluted fuel passed into the recirculation zone in the centre of the swirler to feed a small diffusion flame. In configurations 3 and 4 this pilot flame consumed 12 per cent of the total fuel, but slightly less in configurations 1 and 2. Similar fuel injection arrangements are a feature of some commercial designs. A fifth configuration, designated configuration 3*, in which the swirler was retracted to position 3 but the pilot hole was almost completely blocked, was also investigated.

Fuel and air supplies were measured by rotameters, corrections to the calibrations being applied to account for the pressures in them being greater than atmospheric. An absolute accuracy of ± 2 per cent FSD (full-scale deflection) may be assumed for these flowrates, giving the equivalence ratio typically to about ± 5 per cent. For convenience in discussion, the indicated volumetric flowrate Q_a is used to label the results. With 1.2 l/min of propane, $Q_a = 60$ l/min corresponded to a burner exit velocity of about 4 m/s.

The combustion chamber also had a diameter of 51 mm, and it was a straight pipe, 651 mm long, for these experiments. The bottom section was made of quartz so as to transmit UV radiation efficiently. At the top, the combustion gases were exhausted to the laboratory.

Measurements were made of the chemiluminescent UV emission from excited OH radicals at a wavelength of 307 nm, as discussed in detail in another paper [**16**]. A lens mounted on a telescopic tube directed radiation from specific regions of the flame on to a photo diode mounted behind a UV filter. Stops placed in front of the lens, and the relative positions of the lens and photodiode, determined the dimensions of the collection volume. Three arrangements were employed in the current programme:

Arrangement (a). A 2 mm horizontal slit (with the lens 66 mm and the photodiode 180 mm from the centre of the combustion chamber) collected UV emission from a thin slice of the flame which was 1.1 mm thick at the front and less than 0.1 mm thick at the rear. With a horizontal width varying from 25 mm to 1 mm over the same distance, this collection volume provided a reasonable sample of the total cross-section, and the telescope was traversed vertically to measure the axial distribution of UV emission.

Arrangement (b). A circular stop of 10 mm diameter (with the lens 256 mm and the photodiode 320 mm from the centre of the combustion chamber) led to a

thin waisted collection volume only 1 mm across at front and rear, and about 0.2 mm in the centre. The photodiode was mounted on a pivot so that sweeping the telescope through an angle of 5° allowed the flame to be traversed with nearly parallel lines-of-sight. These signals were deconvoluted using the Abel transform [**16**] to provide profiles of radial emission intensity.

Arrangement (c). This was as in arrangement (b) above, but with the lens 2 mm closer to the centre of the chamber. This moved the conjugate point of the photodiode by 27 mm and broadened the collection width in the centre to 1.3 mm.

The r.m.s. value of the fluctuating photodiode output voltage, suitably amplified, was taken as a satisfactory measure of the instantaneous rate of heat release from the flame in the collection volume, there being a close relationship between the time mean of this type of signal and the r.m.s. value [**16**]. This allowed the signals to be frequency analysed and correlated with the pressure.

Acoustic pressure fluctuations in the combustion chamber at the burner face and in the section of the air supply system just below the tapered section were also measured, using condenser microphones mounted on the ends of short sections of polythene tube. The tubes served to insulate the transducers from both heat and vibration and gave rise to negligible attenuation of the signal. The signals were also amplified and frequency analysed, and their phase, relative to each other and to the UV signals, recorded from a spectrum analyser. This was operated with a bandwidth of 4 Hz over the range 0–800 Hz and the results averaged over 60 samples of 0.25 s.

3 ACOUSTIC ANALYSIS

Typical spectra of acoustic pressure fluctuations in the combustion chamber (p_f), in the supply system downstream of the leaf valve but before the tapered section (p_w), and of the UV emission (b) are shown in Fig. 2. The two pressure transducers had approximately the same calibration; therefore the relative magnitudes of the peaks are approximately correct. Their precise frequency was a function of the burner configuration, of whether the valve was open or nominally shut, and of the fuel, its flowrate and the equivalence ratio. However, the absence of substantial peaks in the combustion chamber acoustic pressure signal, and more particularly in the UV signals, above 400 Hz was a constant feature of this burner at the flowrates examined.

The spectrum of the supply system pressure shows the same peaks as that of the combustion chamber, plus a minor peak at about 500 Hz. An estimate of the

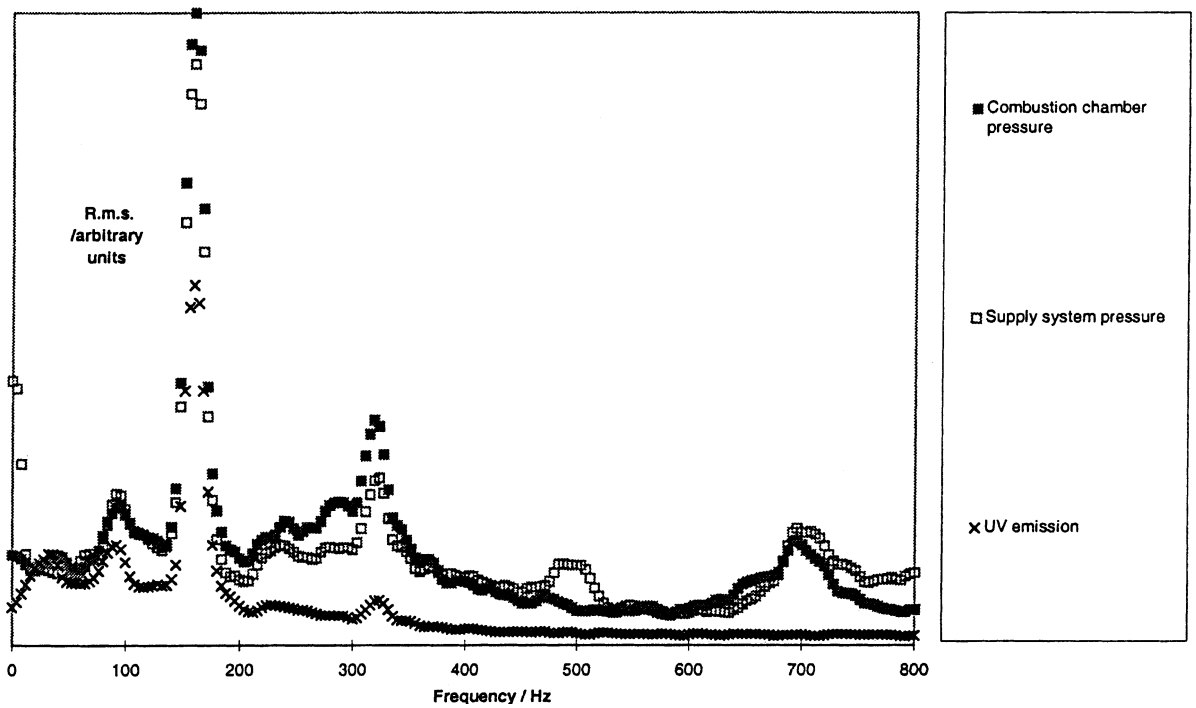


Fig. 2 Typical spectra of acoustic pressures and total UV emission from the flame. The pressures were measured in the flame region of the combustion chamber, and in the supply system just below the swirler and fuel injection 'tree', but above the leaf valve, which was nominally shut. Data are given for methane at an equivalence ratio of 0.59

relationship of the coherent contributions to these spectra is given by

$$|Tr(p_w/p_f)| = \frac{1}{2}\left(\left|\frac{\overline{p_w p_f}}{\overline{p_f^2}}\right| + \left|\frac{\overline{p_w^2}}{\overline{p_w p_f}}\right|\right)$$

At the excited frequencies, these coherent contributions are due to acoustic plane waves. Thus they can be compared with the ratio $|p_w/p_f|$ for these waves and this was calculated over the whole of the spectrum using the methods described by Lawn [**17**]. The burner supply system was modelled as a cylinder of the diameter of the inlet pipe (51 mm), followed by a sudden contraction at the entry to the tapered section. This was taken to have a uniform diameter of 35 mm, which is the mean of the diameters before and after the taper, and a length of 40 mm. The inlet pipe was ascribed a real acoustic impedance at the leaf valve given by KM, where M is the Mach number of the air flow in the pipe. The loss coefficient was measured from steady flow data to be $K = 16$ for the valve open, and $K = 366$ when it was nominally shut. The tapered section, containing the fuel supply tree and the swirler, was measured to have $K = 161$, again based on the full diameter of the inlet pipe.

Comparison of this acoustic model with data taken under three different combustion conditions shows it to predict the correct qualitative behaviour for the magnitude of the transfer function with the valve shut (Fig. 3a) and, more importantly, to yield the correct phase to within about 30° below 400 Hz (Fig. 3b). Some of these data are of low coherence; those with $Co < 0.2$ have been excluded altogether from these plots. A peak in the magnitude of the transfer function just above 500 Hz indicates a Helmholtz resonance, with the entire length of pipe contributing to the resonator volume. Similar comparisons were obtained for the case of the valve open, and in this case the phase predictions were rather more accurate below 400 Hz.

The relevance of these acoustic models is that they enable the acoustic impedance of the burner and its supply system to be evaluated, and hence the acoustic velocity fluctuations u_1 at burner exit relative to the combustion chamber pressure. The impedance at the burner exit plane is

$$Z_1 = \frac{p_f}{\rho u_1 c}$$

where ρ and c are the mean density and speed of sound respectively. The result for the phase with the two valve positions is shown in Fig. 4.

An attempt was made to correlate the principal frequencies of excitation with the Strouhal number based on the axial burner exit velocity U and its diameter d. It was seen that the two lowest frequencies were usually found in the ranges given by $fd/U = 0.4$–0.5 and 0.8–1.0, with sometimes the lower frequency dominant and sometimes the higher. These ranges are quite wide, and the scaling of the frequency on the velocity, even for one burner condition and equivalence ratio, did not hold very precisely. Very often, but not always, the higher frequency was precisely twice the lower frequency.

4 RESULTS

4.1 Spatial distributions of the magnitude of heat release

Most of the qualitative results recorded here from the quantitative UV data were also evident by observation of the (blue) visible spectrum. Initial measurements were made on a methane flame with the valve open, the swirler in the 'original configuration', designated configuration 1 in Section 2, and with the UV collection arrangement designated arrangement (b) in Section 3. Taking the full spectrum of fluctuations into account, the evolution of the radial UV intensity distributions with axial distance at $\phi = 0.42$ was as shown in Fig. 5a. The flame spreads out from the region surrounding the pilot flame at $x = 5$ mm, the UV intensity reaches a peak near the axis between 15 and 20 mm, and then the profile flattens and decays.

The result of seating the swirler to produce burner configuration 2, and closing the valve, can be seen by comparing Fig. 5b with Fig. 5a. In Fig. 5b the contributions above and below 300 Hz have been separated at two axial distances but, since these contributions have similar distributions in both cases, comparison with Fig. 5a is valid. Clearly, eliminating the annulus of high-speed unswirled air around the swirler has produced a much hollower flame, with virtually no heat release on the axis above the pilot flame and considerable combustion taking place in the outer region at an axial distance from the burner outlet plane of $x = 25$ mm. While the deconvolution analysis makes the radius of cut-off of the flame ($r = 20$ mm or 4°) at $x = 10$ mm sharper than it should be, there was in fact very little emission outside this position, reflecting the 'mushroom' shape of the flame.

The same results for $x = 10$ mm are shown in Fig. 5c, but there they are compared with some for a higher equivalence ratio. This shows that raising the premixture to $\phi = 0.59$ by reducing the air flow for the same fuel flow has the effect of initiating combustion in the premixture around the pilot flame much earlier.

With the swirler retracted into configuration 3, further results for methane were then built up from radial traverses with optical arrangement (c) and deconvolution at nine different axial distances for a specific self-excitation frequency (Fig. 6a). (The contribution of the pilot flame to the heat release, below $x = 7.5$ mm, is exaggerated here by the increased UV emission associated with diffusion burning at stoichiometric condi-

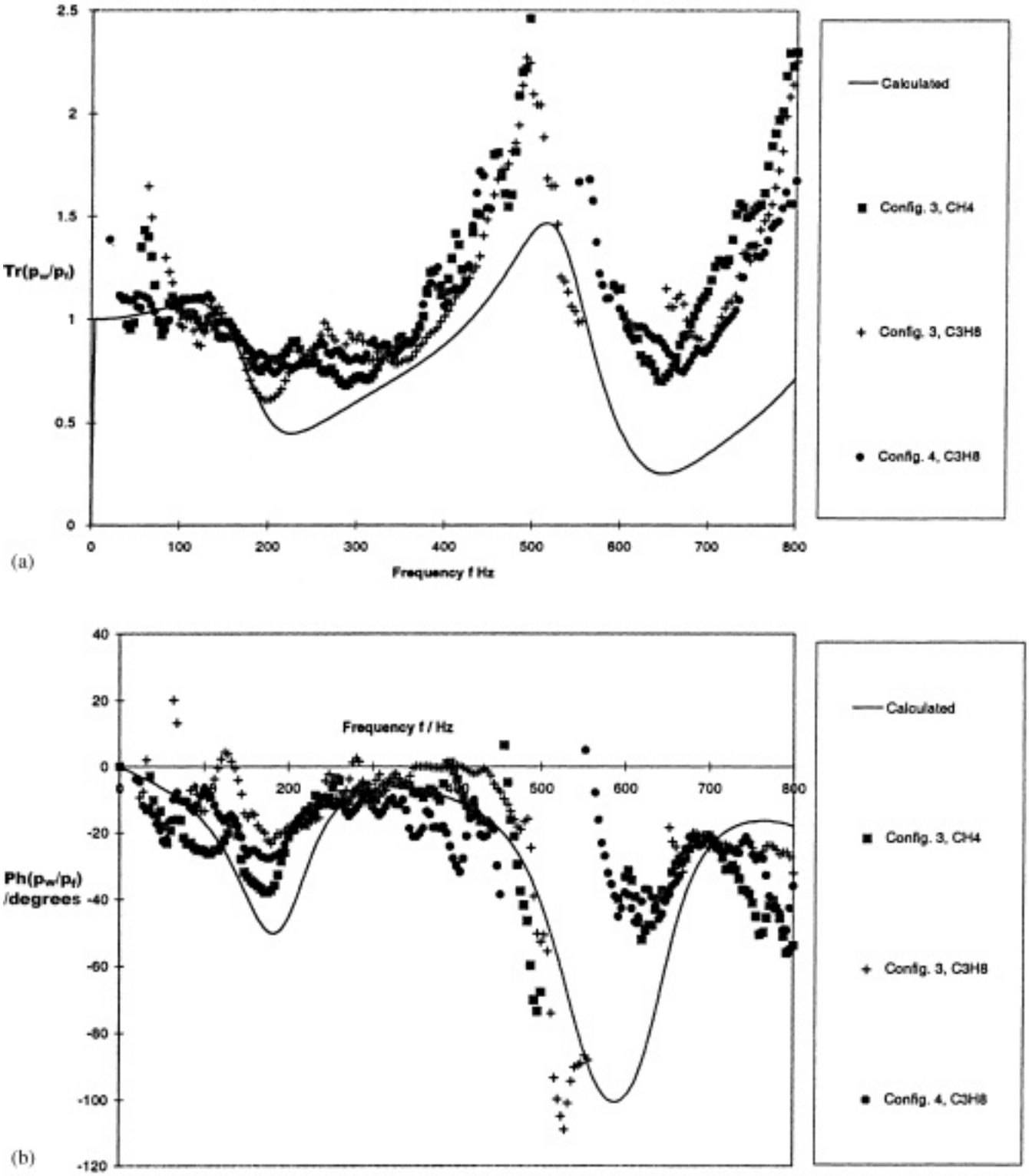


Fig. 3 (a) Magnitude and (b) phase of the transfer function between the supply system and combustion chamber pressures with the leaf valve nominally shut. Data of coherence greater than 0.2 are compared with the plane-wave theory, the excitation being by flames of methane and propane

tions, compared with that in the premixture in the bulk of the flame.) The peak contribution of the premixed flame to the fluctuating heat release is at a distance of about 20 mm, and off the axis, but not as far off as with the swirler seated. The profile for $x = 15$ mm at this frequency is in fact rather similar to that in the lower-

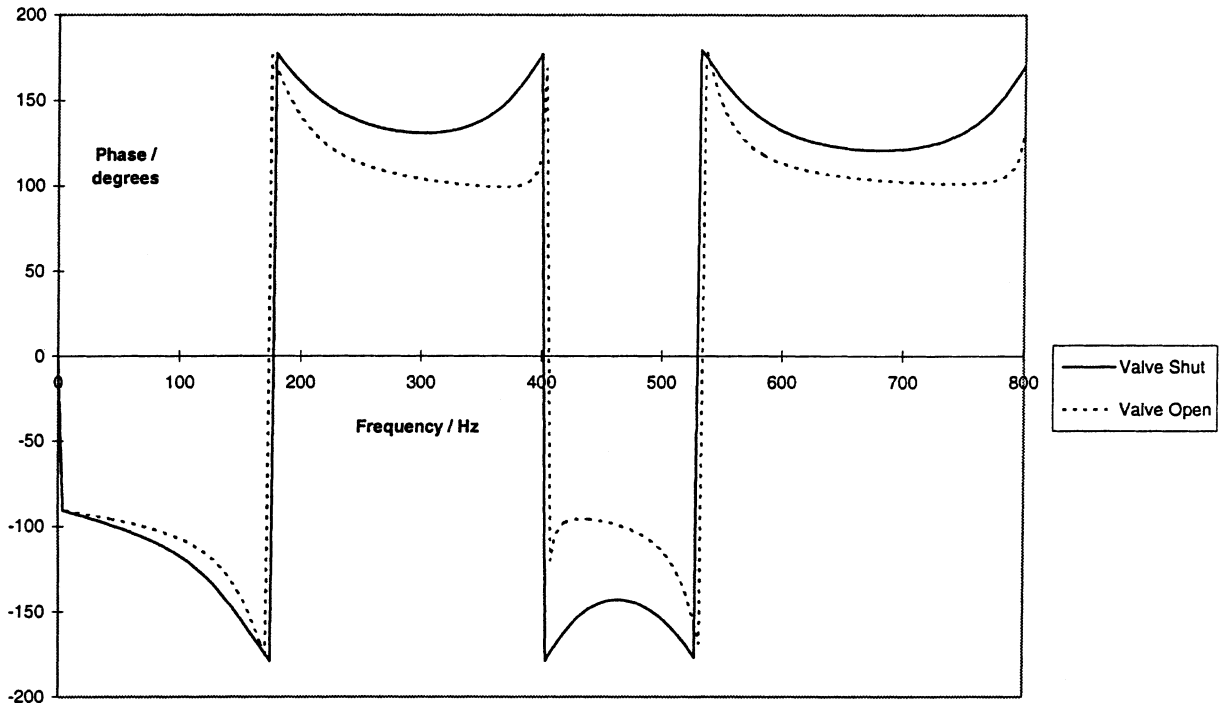


Fig. 4 Theoretical variation in the phase of the acoustic velocity at the inlet to the combustion chamber relative to the pressure, for the two extreme positions of the leaf valve in the air supply system

frequency band for $x = 10$ mm with the swirler seated. Thus the general effect of retracting the swirler is to delay the heat release at the excited frequencies.

With the same retracted swirler configuration 3 but with the burner operated on propane and $\phi = 0.41$ and 0.31 (Fig. 5d), the distributions are very similar to those for methane at $\phi = 0.42$ with configuration 2 (Fig. 5b) for both cases, with little combustion on the axis. At $x = 10$ mm, the radially restricted flame is again evident. Thus, the effect of changing from methane to propane is counteracted by retracting the swirler.

4.2 Spatial distributions of the phase of heat release

The phases of the UV fluctuations relative to the combustion chamber pressure at the same excited frequency for methane are shown in Fig. 6b. These phases are as recorded along chordal lines of sight and hence are not strictly those at the local position shown in the plot. However, a deconvolution of the real and imaginary components of the signal for one axial position in which there was a substantial variation ($x = 5$ mm) showed that the error involved in ascribing the phases to the closest approach of the line-of-sight to the axis, as here, is likely to be small.

It is seen that the outer regions of the flame are approximately in phase and lagging the pressure by 100–120°. The pilot flame lags by slightly more but, in the region of greatest magnitude, the UV emission is almost in phase with the pressure. Further along the flame, the phase lag increases steadily.

4.3 Axial distributions of the phase of heat release

Precisely how the phase lag increases was investigated by measuring axial distributions using the UV arrangement (a). The results are displayed in Fig. 7 relative to the axial distance x, and in Fig. 8 relative to $f(x - x_0)/U$. This is the distance from an effective origin x_0 (determined as described below), normalized by the distance for convection at the burner exit velocity U during 1 cycle. In both cases, the phases displayed are not relative to the pressure, but to the burner exit velocity fluctuation u_1, the conversion being through the data of Fig. 4. (The small changes in phase due to changes in the mean flowrate were ignored in this calculation.)

The data for methane are those in Fig. 7a. Most of the data are for $\phi \approx 0.6$ and configuration 1. The degree of repeatability, the effect of increasing the air and fuel velocities, and the effect of changing from configurations 1 and 3 are shown. A second harmonic was recorded for one of these runs, and this too is plotted. The final data set is for the equivalence ratio reduced to 0.46.

The data for the obvious linear decrease in phase in the region of high heat release were extrapolated with best-fit lines to establish an effective origin where there would be coincidence between the disturbances and the velocity

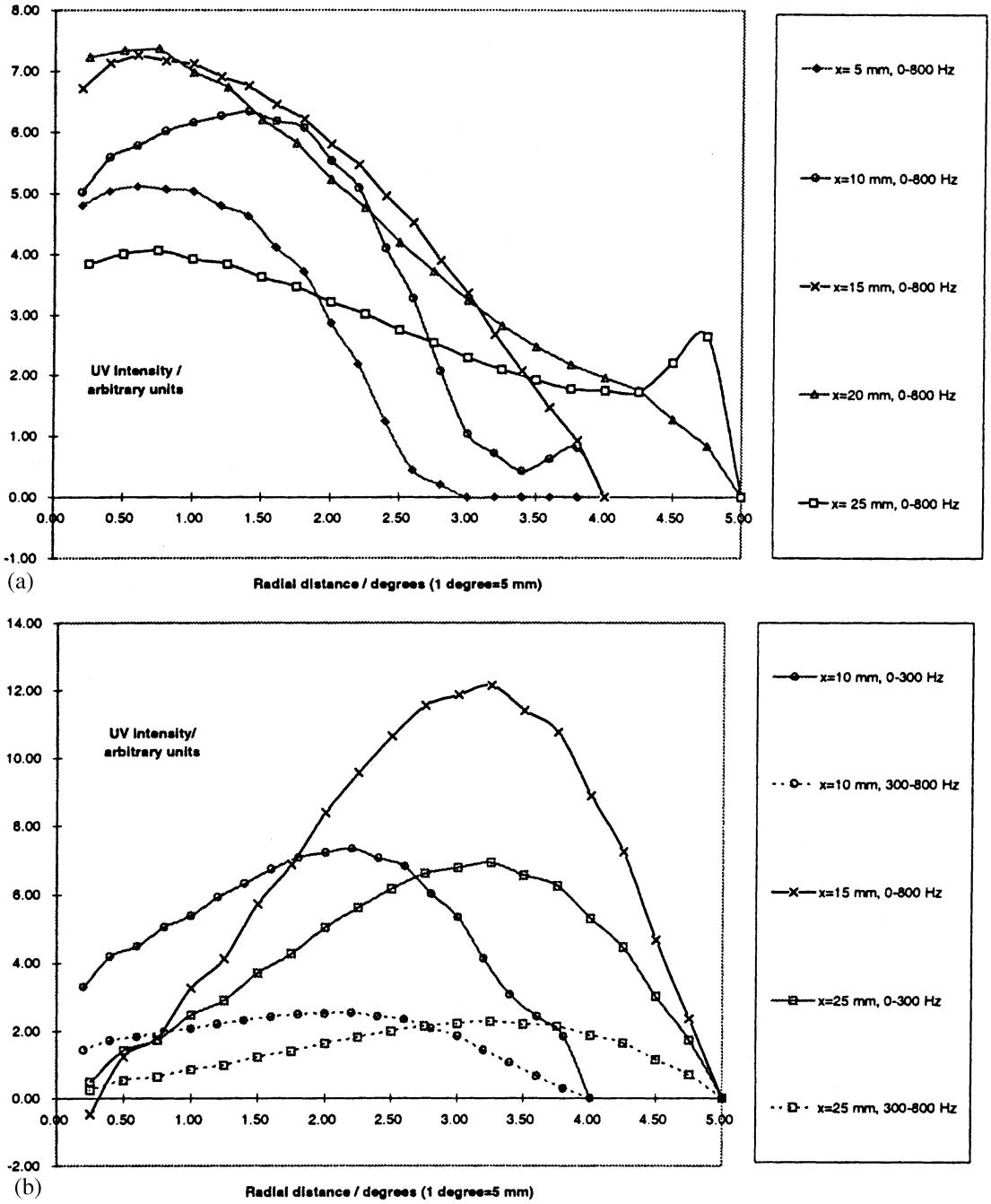


Fig. 5 (continued over)

fluctuations if the convection velocity was constant in the axial direction. It is seen that, for the high equivalence ratio, these origins lie in the x_0 range from -14 to -18 mm, in spite of the wide range of frequencies and the variation in velocity. When plotted against the dimensionless ratio discussed above with $x_0 = -14$ mm, the data effectively collapse on to two parallel lines, indicating a common convection velocity which scales on the burner exit velocity (Fig. 8a). The region of the pilot flame, before the region of high heat release, appears to be rather more than 180° out of phase with the main flame.

With the burner operated on propane in configuration 3, a much larger number of conditions were investigated, involving changes to air and fuel flows, and hence equivalence ratios (Fig. 7b). In a number of instances, the second prominent frequency was recorded, and this was often of greater magnitude in the pressure and UV spectra than the first. A new effective origin at $x = -5 \pm 4$ mm for all but one out of eight runs with $\phi > 0.43$ was

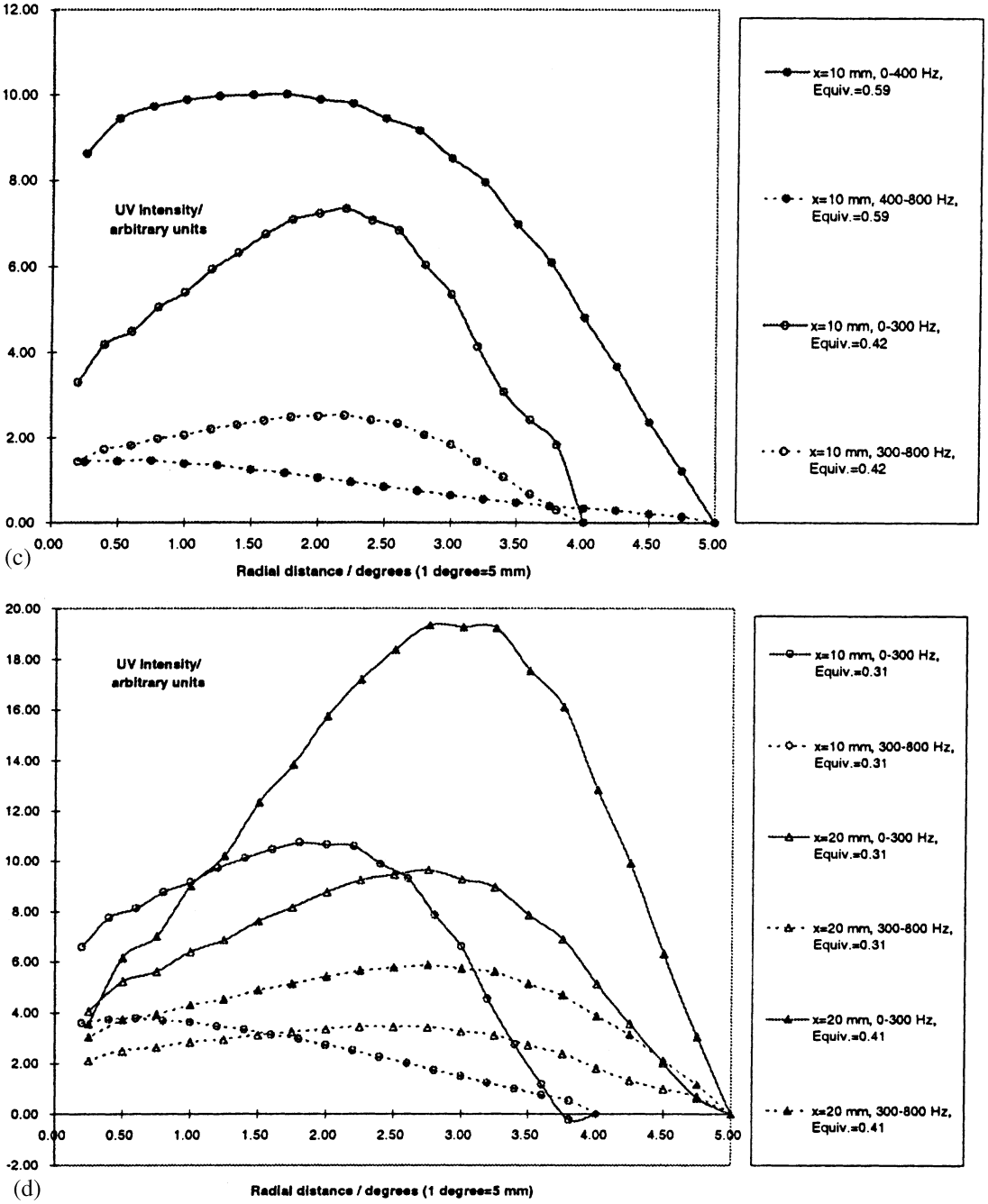


Fig. 5 Radial distribution of UV intensity in the specified frequency ranges, deduced by deconvolution of the line-of-sight measurements, at various heights: (a) methane in configuration 1, $\phi = 0.42$; (b) methane in configuration 2, $\phi = 0.42$; (c) methane in configuration 2, $\phi = 0.42$ and 0.59; (d) propane in configuration 3, $\phi = 0.31$ and 0.41

identified. (The anomaly is not plotted here.) Then, rationalizing these data in the same way as above, the phase lag was seen to correlate well with the methane data (Fig. 8b), indicating the same convection velocity relative to the burner exit velocity.

Some of the low-frequency modes are now found to involve a continuation of the phase relationship which is associated with the pilot flame region in the higher frequencies. This is a lag of about 230° relative to the main region. Between the two distinct regions, there was always a transition region in which the correlation between UV and pressure virtually disappeared. This was probably because one mode was present in the central pilot flame and the other, of similar magnitude, in a surrounding premixed sheath, as indicated by Fig. 6b.

Two of the data sets for this configuration 3 are compared in Figs 7c and 8c with results from configuration 3*, in which the pilot flame hole was blocked. A

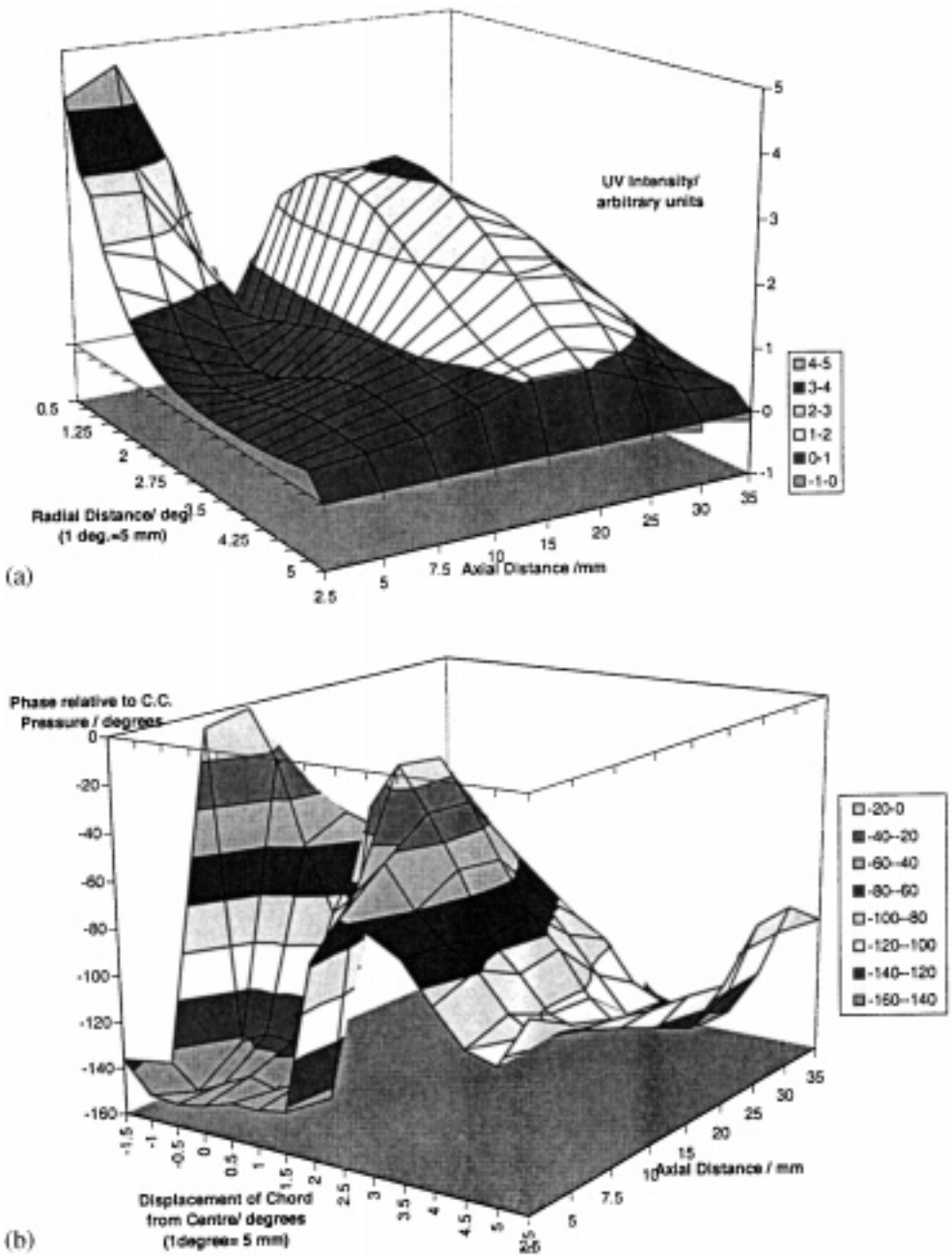


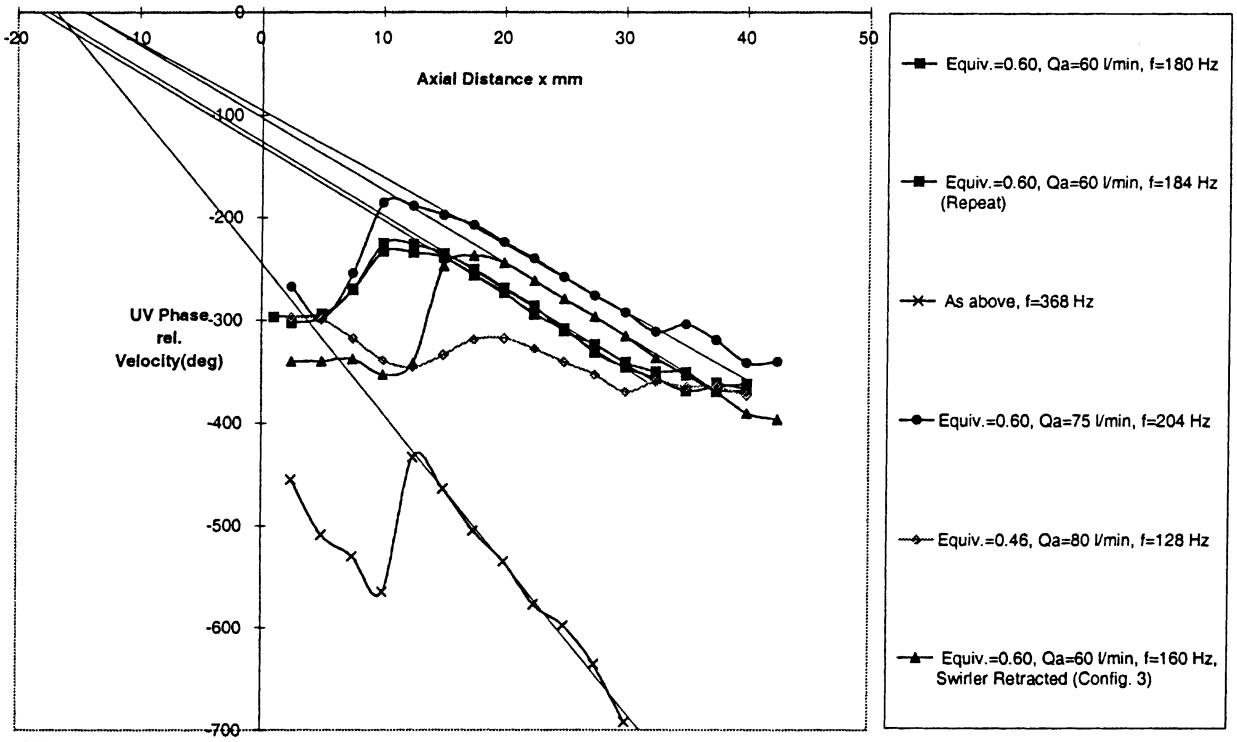
Fig. 6 Axial and radial distributions of (a) the magnitude and (b) the phase (measured along chords) of the UV intensity relative to the combustion chamber pressure at a frequency of 160 Hz for methane in configuration 3 and $\phi = 0.59$

completely different effective origin was located, at $x_0 = -35$ mm, and the initial region disappeared. However, the same gradient is observed for the main region on the normalized plot (Fig. 8c).

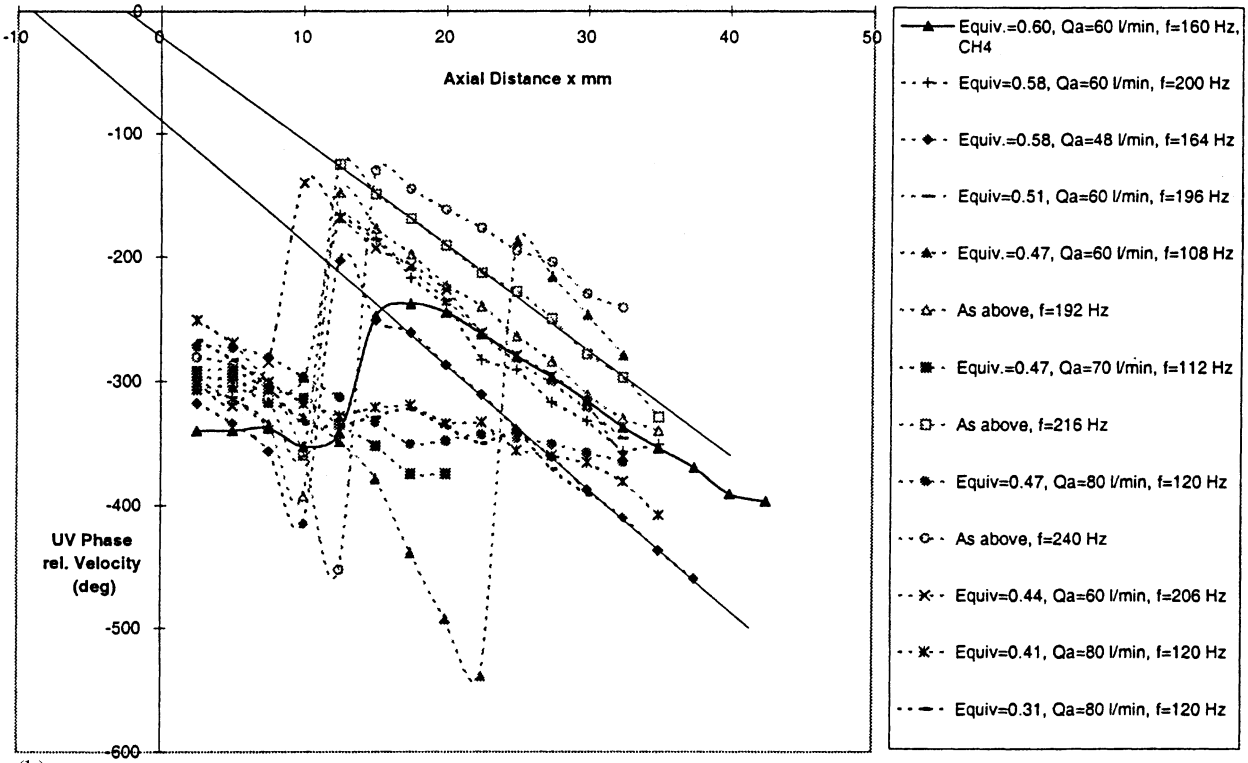
Finally, the effect of increasing the swirler retraction is examined in Figs 7d and 8d, in which results for configurations 2, 3 and 4 are compared. The configuration 4 results for the swirler further retracted indicate an effective origin at $x_0 = -18$ mm. For configuration 2, with the swirler seated, extrapolation to the effective origin with accuracy was difficult, although a value of $x_0 = -14$ mm was the best estimate. Very different convection velocities are observed for the three configurations.

4.4 Axial distributions of the magnitude of heat release

The axial distributions of UV intensity at the most



(a)

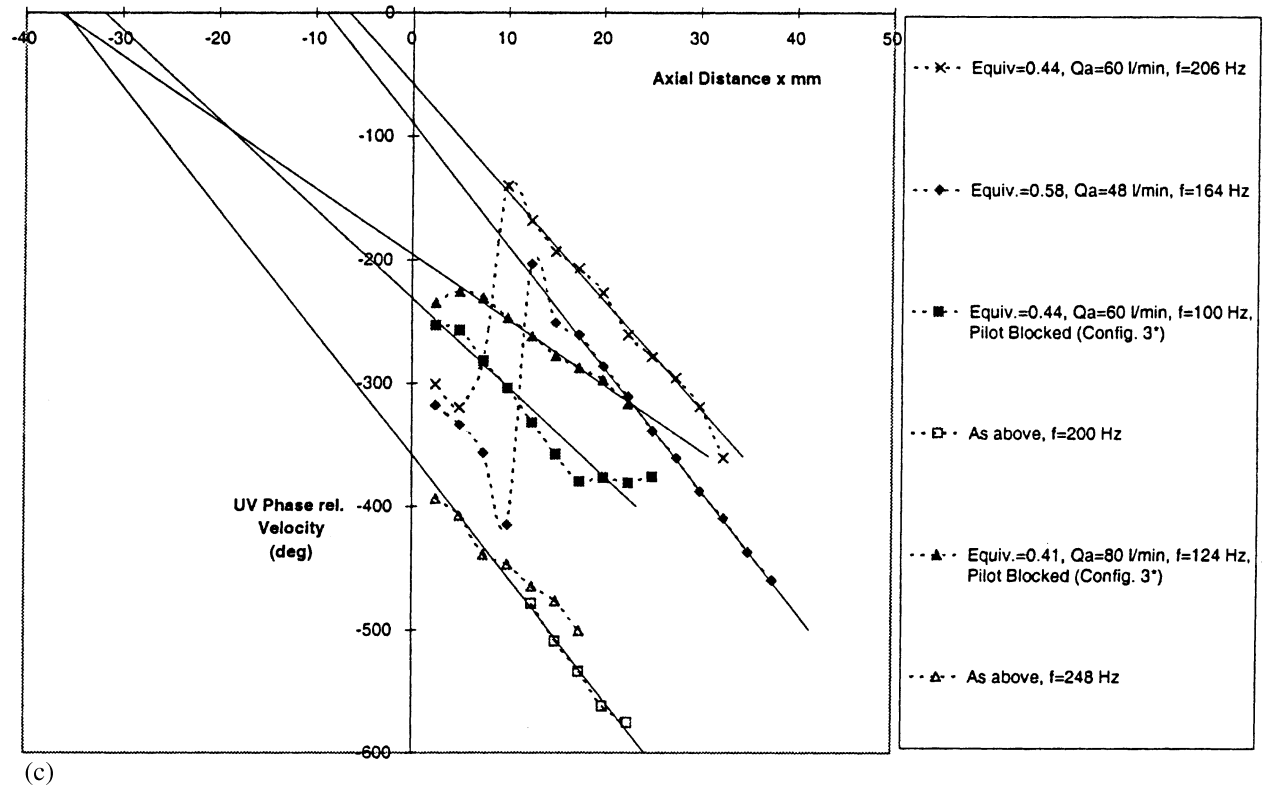


(b)

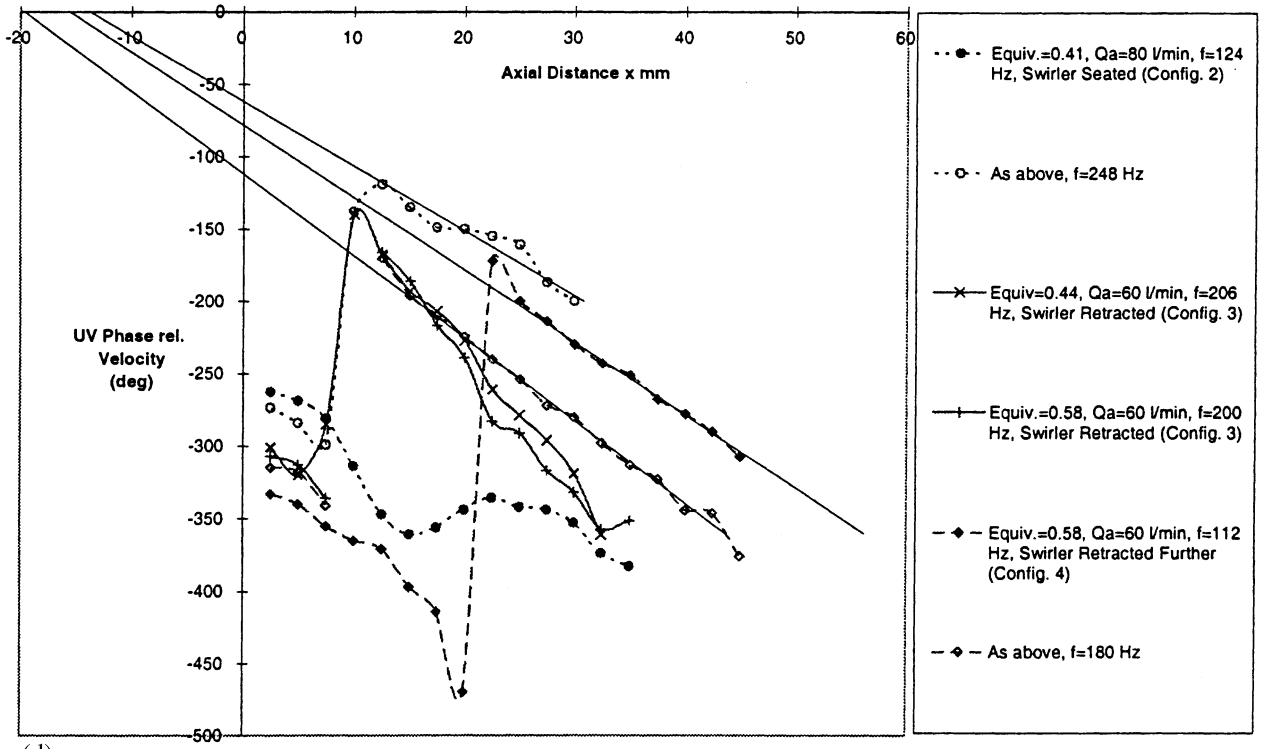
Fig. 7 (continued over)

prominent frequencies for each condition were measured with the optical arrangement (a) at the same time as the phase. The units of intensity are arbitrary but consistent for the same equivalence ratio. However, the change in calibration with ϕ meant that there was no additional penalty for scaling the intensity signals for comparative purposes.

As shown in Fig. 9a for methane burnt in configuration



(c)



(d)

Fig. 7 Axial distribution of the phase of the heat release relative to the burner velocity: (a) for methane burning in configuration 1, except where noted as configuration 3; (b) for propane burning in configuration 3, except where noted as methane (this shows the effect of the equivalence ratio and fuel); (c) for propane burning in configuration 3, except where noted as having the pilot blocked (configuration 3*); (d) for propane burning in configurations 2, 3 and 4 (this shows the effect of swirler retraction)

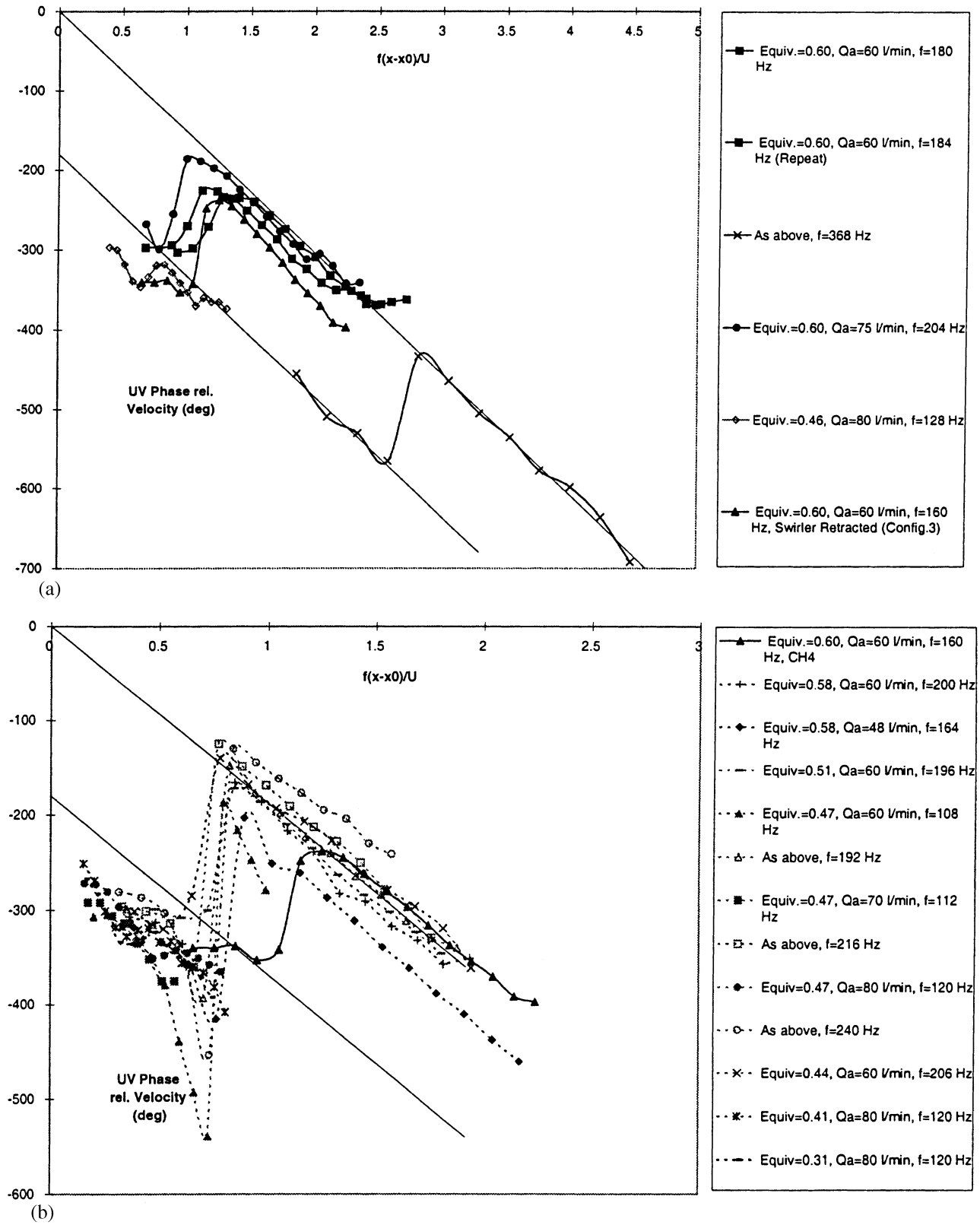
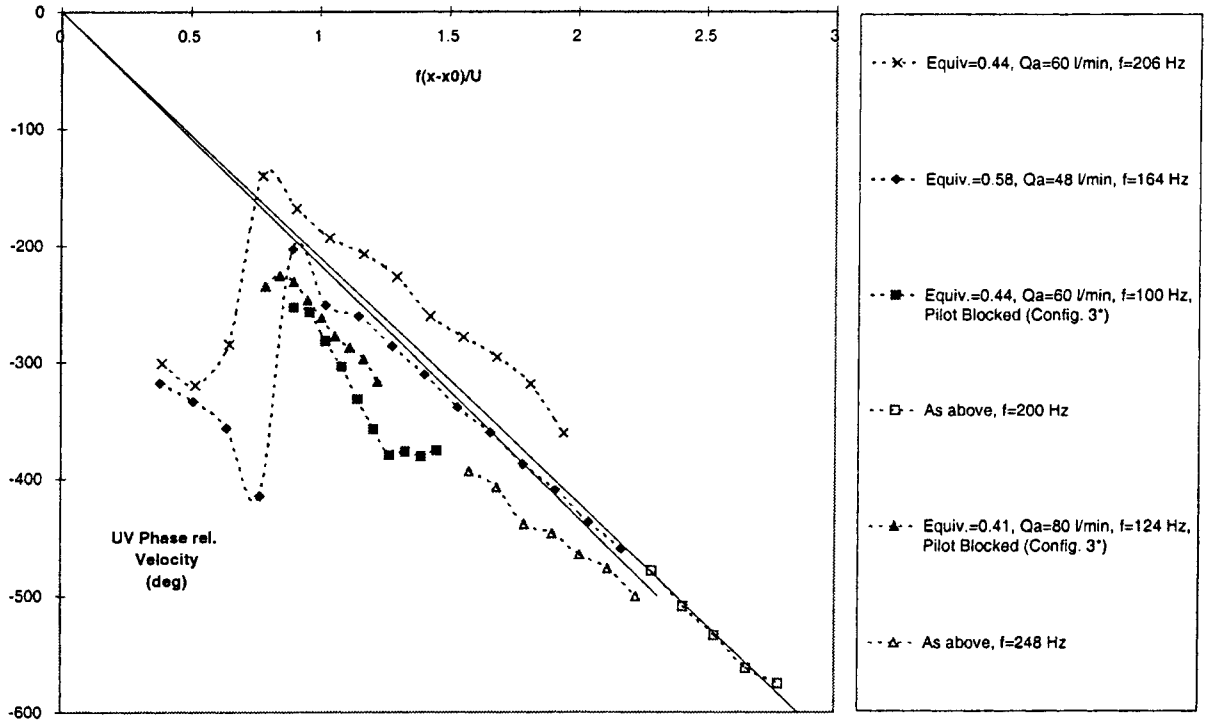


Fig. 8 (continued over)

1 (corresponding to Figs 7a and 8a), the repeatability of these data was also good, although the one result for a higher mass flowrate but the same equivalence ratio (0.6) suggested a slight lengthening of the spatial distribution (by about 3 mm). A lower ϕ (0.46) led to suppression of the heat release fluctuation in the main region, leaving the



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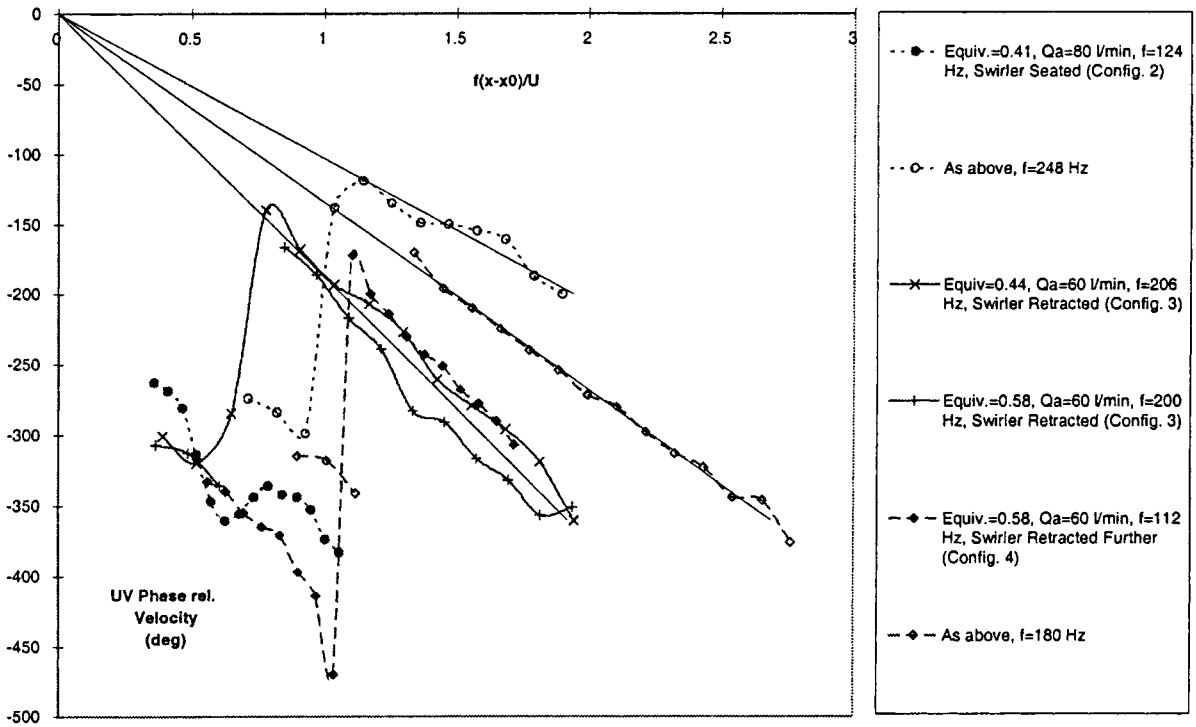
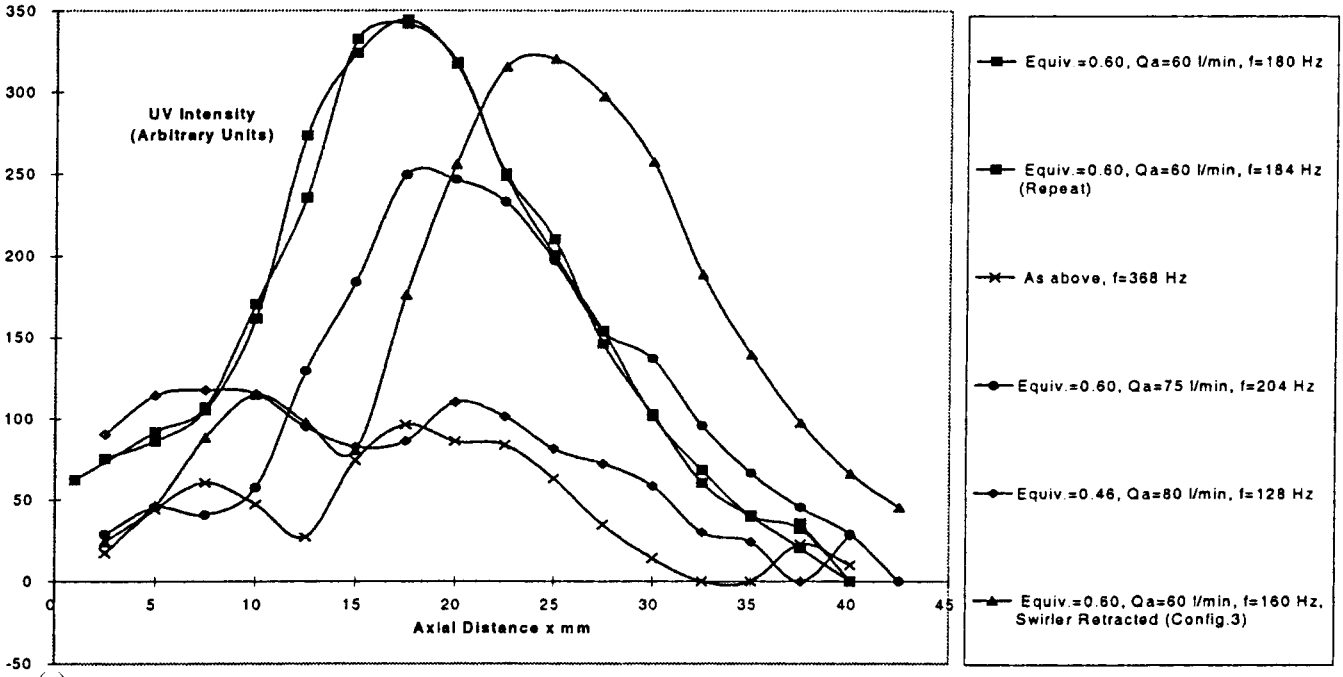


(d)

Fig. 8 Distribution of the phase of the heat release relative to the burner velocity as a function of the normalized axial distance: (a) for methane burning in configuration 1, except where noted as configuration 3, with a common effective origin $x_0 = -14$ mm; (b) for propane burning in configuration 3, except where noted as methane, with a common effective origin $x_0 = -5$ mm for propane (this shows the effect of the equivalence ratio and fuel); (c) for propane burning in configuration 3, except where noted as having the pilot blocked (configuration 3*), with an effective origin $x_0 = -5$ mm for pilot open and $x_0 = -34$ mm for pilot blocked; (d) for propane burning in configurations 2, 3 and 4, with an effective origin $x_0 = -14$ mm for the swirler seated (configuration 2), $x_0 = -5$ mm for the swirler retracted (configuration 3) and $x_0 = -18$ mm for the swirler retracted further (configuration 4) (this shows the effect of swirler retraction)



(a)

Fig. 9 (continued over)

pilot flame prominent in the initial region. A slightly more compressed distribution was evident at a higher frequency than at a lower frequency for $\phi = 0.6$. Retracting the swirler to configuration 3 had the effect of delaying the fluctuating heat release by about 7 mm.

With propane burnt in configuration 3, the lower- and higher-frequency modes, as distinguished by their Strouhal numbers, have been separated for clarity in presentation. At lower frequencies [Fig. 9b(i)], the similarity of the distributions for the same ϕ but different velocities is clearly demonstrated, negating the possibility of correlation with the transit time x/U. The major contributions come from near the burner exit, and they are suppressed when ϕ is reduced or the fuel is changed to methane. At higher frequencies [Fig. 9b(ii)], there is evidence of a longer response as ϕ is reduced. The lowest value of $\phi (= 0.44)$ for which an axial traverse was undertaken and which led to a high intensity of fluctuation inexplicably reverses this trend.

Two of these data sets are shown again in Fig. 9c, where the effect of blocking the pilot jet with the swirler in configuration 3 is seen to be quite dramatic in shortening the responsive part of the flame. This is consistent with the very substantial movement of the effective origin into the body of the burner. The lower-frequency mode is again dominant in the near-burner region and is therefore not dependent on there being a pilot flame. With the pilot in operation, seating the swirler also shortens the response, while retracting it further clearly makes it much longer (Fig. 9d), in both high- and low-frequency modes.

5 DISCUSSION

5.1 The effect of varying the excitation frequency

For most of the data presented here, there was little difference between the radial distributions of UV intensity in the lower-frequency band (less than 300 or 400 Hz) in which the acoustic excitation took place, and those in the higher-frequency band associated with the background turbulence [**16**]. This was also true of the frequency-integrated axial distributions, although these have not been presented. However, in Figs 5c and 5d there is evidence of a tendency for the integral over the acoustically excited frequencies to be relatively smaller on the axis than that over the higher frequencies. Moreover, the lower-frequency mode ($fd/U < 0.5$) clearly had a different axial distribution from the higher-frequency mode (Fig. 9). Both of these modes had $f < 400$ Hz and were thus in the lower-frequency band, but different mechanisms of excitation are indicated.

Changing the frequency of excitation by opening the air supply valve while maintaining the flowrate and equivalence ratio made only a small difference to the axial distributions. If the time-averaged heat release

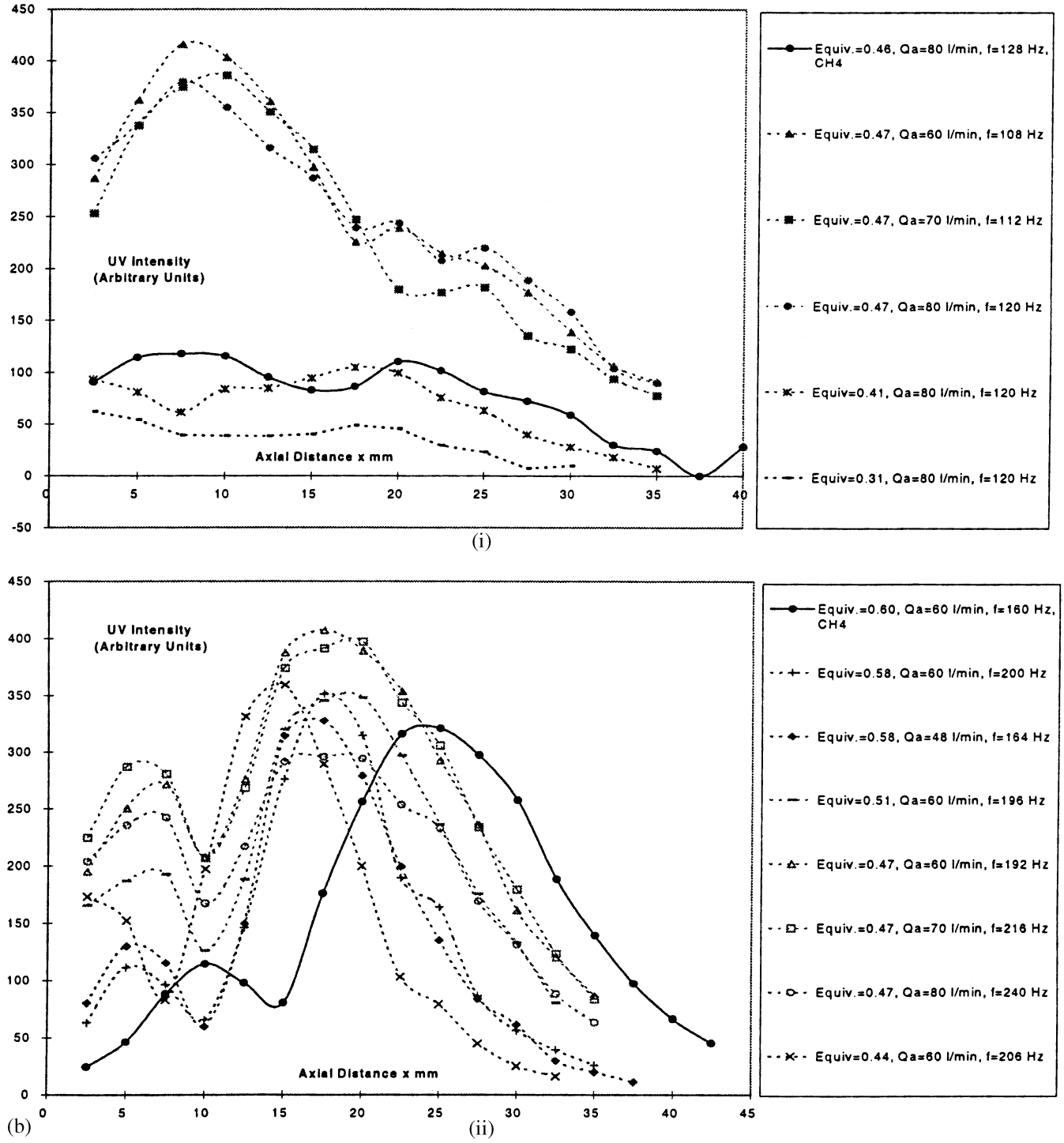
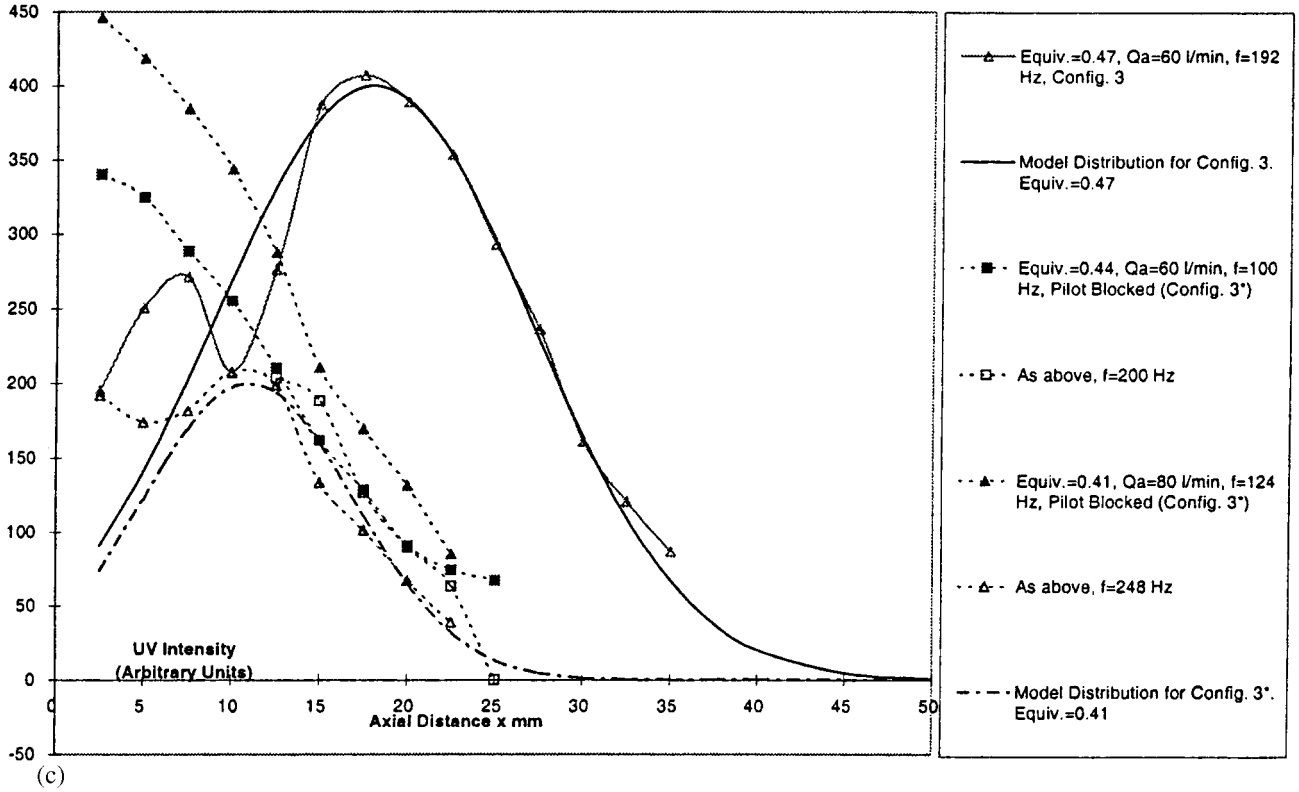


Fig. 9 (continued over)

distribution is unchanged by opening the valve, no difference at all would be expected, because the local heat release would simply distribute itself in a different way around the spectrum in the lower-frequency band (less than 400 Hz), not affecting the integral over that band, or the integral over higher frequencies. The r.m.s. value of UV emission at discrete frequencies can be regarded as a measure of the heat release perturbation generated by the acoustic wave at that frequency, and the integrals as measures of the time-averaged heat release.

These observations indicate that the perturbations to the flow and the heat release associated with the higher-frequency mode were small, superimposed on the normal



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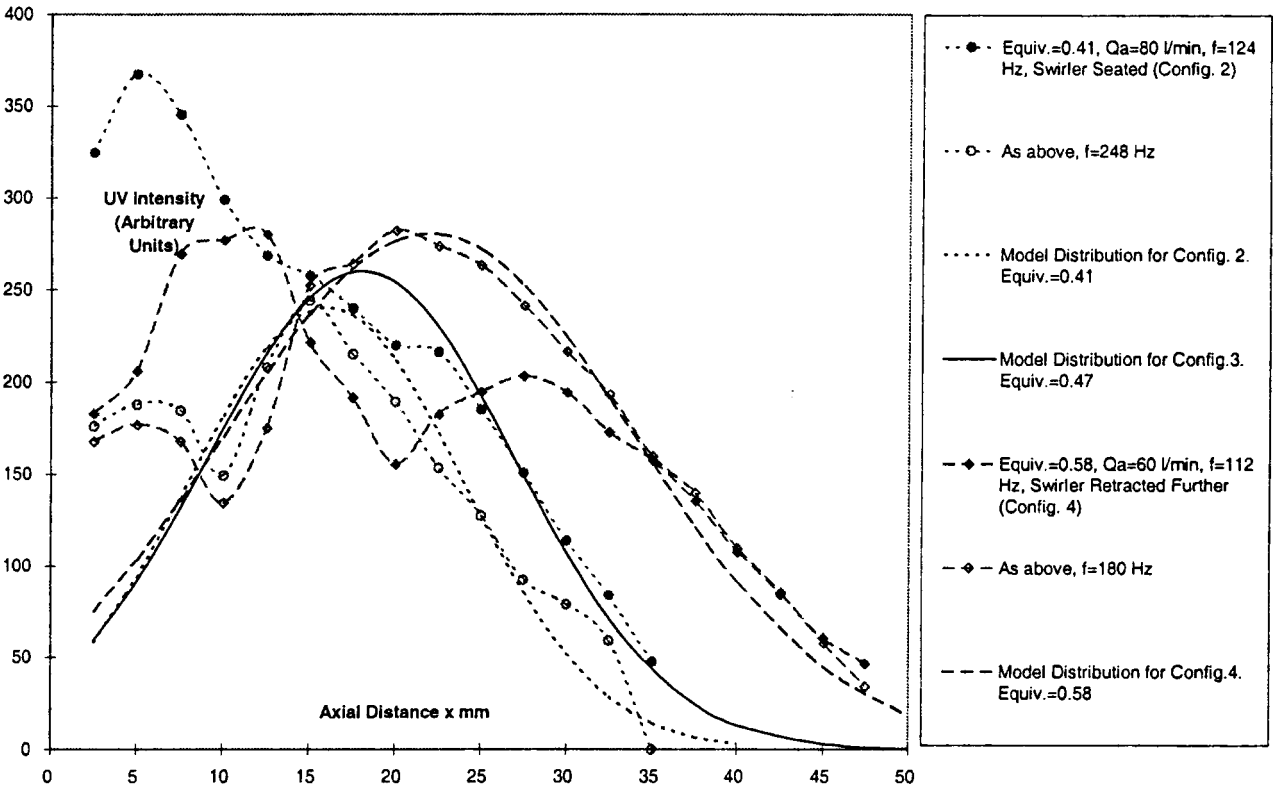


(d)

Fig. 9 Axial distribution of the magnitude of the heat release: (a) for methane burning in configuration 1, except where noted as configuration 3; (b) for propane burning in configuration 3, except where noted as methane in configuration 1, (i) for the lower-frequency mode and (ii) the higher-frequency mode (this shows the effect of the equivalence ratio and fuel); (c) for propane burning in configuration 3, except where noted as having the pilot blocked (configuration 3*); (d) for propane burning in configurations 2, 3 and 4 (this shows the effect of the swirler retraction)

turbulence and probably in the outer regions of the flame downstream of the pilot. (Radical changes to the flame shape were visibly generated at higher equivalence ratios than these, with higher resulting acoustic pressures.) For the data presented, an approximate calibration of the pressure transducer indicates r.m.s. pressures of the order of only 7 Pa. With the calculated exit impedance for the premixture supply system, this translates to $\tilde{u}_1 = 0.2$ m/s, or $\tilde{u}_1/U \approx 0.03$, consistent with the first-order perturbation view.

On the other hand, the lower-frequency mode involved a completely different axial distribution, with some contributions on the axis, and a phase relationship which suggested a continuation of the pilot flame behaviour. For the higher-frequency mode, the complex results for the axial distribution can be summarized with the aid of a simplified model which goes some way towards explaining the conditions under which certain flames are liable to produce excitation.

5.2 An empirical model for the heat release

With no attempt to characterize the near-burner region, an empirical expression was fitted to the magnitude of heat release q for the higher-frequency mode in the main region:

$$\int |q| \, \mathrm{d}A \propto \exp\left[-\frac{(x-x_1)^2}{2s^2}\right]$$

with the peak position x_1 and the standard deviation s of the Gaussian distribution chosen to give the best fit. Examples of these fits are shown in Figs 9c and d. Furthermore, as already explored, the phase lag relative to the acoustic velocity can be expressed as

$$\phi_{q/u} = -2\pi \frac{U}{U_{\mathrm{eff}}} \frac{f(x-x_0)}{U}$$

the slope of the best-fit lines in Fig. 8 determining U_{eff}/U and hence the effective convection velocity. Thus the Rayleigh integral can be written in terms of the four burner parameters x_0, x_1, s and U_{eff}, the frequency and the phase of the burner acoustic velocity relative to the pressure, $\phi_{u/p}(f)$, which is determined by the supply system acoustics:

$$\int_0^\infty \int |p||q| \cos \phi_{q/p} \, \mathrm{d}x \, \mathrm{d}A \propto \int_0^\infty \exp\left[-\frac{(x-x_1)^2}{2s^2}\right] \times \cos\left(\phi_{u/p} - \frac{2\pi U}{U_{\mathrm{eff}}} \frac{fd}{U} \frac{x-x_0}{d}\right) \mathrm{d}x$$

The burner parameters determined by fitting the data in Figs 8 and 9 are given in Table 1, which is therefore a summary of the experimental results for phase and magnitude in a simplified form.

5.3 Explanation for the different flame parameters

It is possible to explain some of these observations for the higher-frequency modes in terms of the overall pattern of combustion. Although the pilot flame functions in a way which anchors the main flame and broadens the limits of flammability, it appears from comparing configurations 3 and 3* in Fig. 9c that it delays the combustion in the main zone. Certainly, the premixed flame could be seen to burn back into the burner exit when there was no pilot.

The effect of reducing the equivalence ratio appears from most of the data for propane in Fig. 9b, and from those for methane in Fig. 5c, to be to lengthen the flame. This is in agreement with the results of computation and observation on a swirled flame by Bradley *et al.*[**18**], who explained it in terms of the quenching of the flame in the shear layer when the laminar burning velocity is small, as in lean premixtures. The fit of the model equation to the data for $\phi = 0.47$ and 0.58 is shown in Figs 9c and d.

As the swirler is retracted progressively, the flame also lengthens (Figs 9a and d), coming closer to the axis (Figs 5a and b). The high-velocity unswirled premixture around the periphery of the swirler is carried further downstream before it can ignite than when all the flow is swirled.

With the pilot in operation, the effective origin for disturbances in the main flame shows remarkable consistency at $x_0 = -16 \pm 2$ mm, a position close to the

Table 1 Summary of the fitted data in the main region of the flame for the higher-frequency modes

Fuel	Valve	Swirler	Configuration	Equivalence ratio	x_1 (mm)	s (mm)	x_0 (mm)	U_{eff}/U
CH_4	Open	Original	1	0.60	18	7	−17	2.3
CH_4	Shut	Retracted	3	0.60	25	7	−14	2.0
C_3H_8	Shut	Seated	2	0.41	16	8	−14	3.5
C_3H_8	Shut	Retracted	3	0.47	18	9	−5	1.9
C_3H_8	Shut	Retracted	3	0.58	18	5	−5	1.9
C_3H_8	Shut	Retracted; no pilot	3*	0.41	11	6	−34	1.7
C_3H_8	Shut	Further retracted	4	0.58	22	12	−18	2.7

branches of the fuel injection tree, for all conditions but those for propane with the swirler retracted to configuration 3. The shift in this position to $x_0 = -5 \pm 4$ mm with this configuration and $\phi > 0.43$ is not easy to explain.

This apparent maximum in the approach of the effective origin to the outlet is, however, consistent with the deduced convection velocities, which show a minimum in this configuration. These velocities are influenced by the uniformity of the exit flow, and by the balance between the spreading out of the swirled flow and the rate of temperature rise and acceleration due to combustion. It may be conjectured that in configuration 2 the temperature effect is dominant and the convection more rapid than in configuration 3, while in configuration 4 the velocities of burning gas are higher due to the unrestricted annular flow and to the reduced flame spread.

The convection velocities are 2.0 ± 0.3 times the nominal exit velocity for all the configurations in which there was a modest swirler retraction, increasing to 2.7 due to the higher proportion of unswirled premixture and combustion near the axis with further retraction. When the swirler was seated, however, the earlier expansion also gave rise to an increased average convection velocity, of 3.5 U. Observations on a Bunsen flame by Buechner *et al.* [**4**] apparently needed no such correction to the exit velocity, while those on an annular jet showed convection to be slightly slower than the mean burner exit velocity [**15**].

The possibility that some of these modes were in fact helical, as observed by Paschereit *et al.* [**14**], was specifically investigated by looking for significant phase shifts between perpendicular UV lines of sight. None was found.

Finally, Fig. 9bii indicates that the lower laminar burning velocity for methane than propane at $\phi = 0.6$ (0.07 as opposed to 0.13 m/s [**19**]) also results in a longer flame. Jones and Leng [**20**] found significant differences in the behaviour of a pulse combustor with just 20 per cent additions of propane to methane. However, the possibility of relating the results for the two fuels at different equivalence ratios through the laminar burning velocity seems to be eliminated by the fact that the methane flame was stabilized well below the normal limit of lean flammability ($\phi = 0.5$) by the pilot.

5.4 Conditions for excitation

For each of these flame conditions, the contribution to the Rayleigh integral was evaluated as a function of frequency at each axial location. Similar data were also generated directly from the measurements; these showed the same trends and that the contributions from the pilot flame were small. These model data are presented in Fig. 10a in order to illustrate the mechanism for producing a positive value of the integral more clearly. At the chosen frequency of 272 Hz and a velocity corresponding to $Q_a = 80$ l/min, both the swirler-seated (configuration 2) and swirler-retracted (configuration 3) cases have positive integrals. However, the negative contribution arising from the convection time delay in association with the later heat release due to retracting the swirler further (configuration 4) prevents a positive integral. Similarly, the reduction in Q_a to 60 l/min with configuration 3 also reduces the net energy input at this frequency because the smaller convection delay puts the peak heat release out of phase with the pressure. Broda *et al.* [**13**] have produced a complete map of the contributions to the Rayleigh integral for one combustion condition, and then integrated the data to yield an axial variation similar to that of configuration 2.

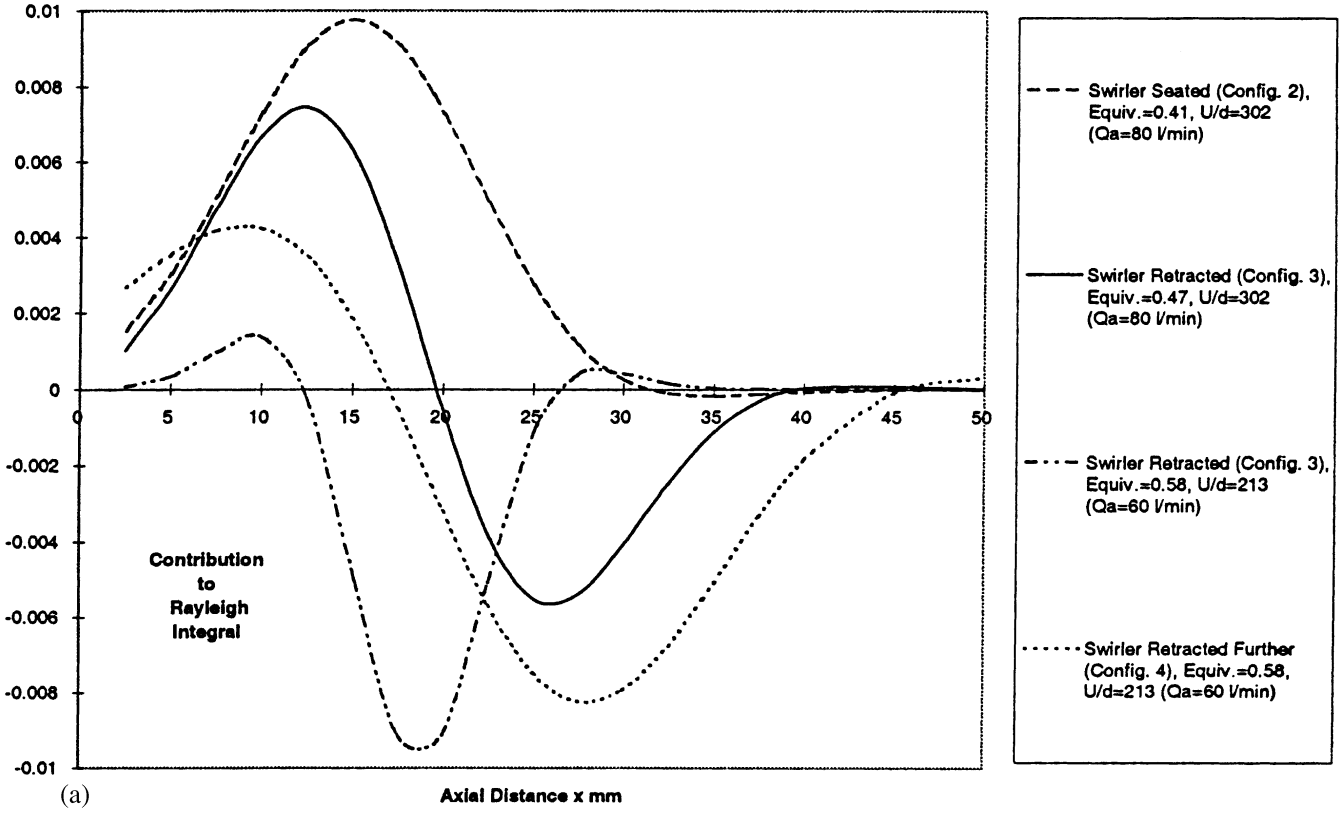
These mechanisms lead to the complex variation in the Rayleigh integral with frequency illustrated in Fig. 10b. Excitation is possible in the bands where the integral of these curves is sufficiently positive to overcome the acoustic dissipation, although its realization depends on satisfying the phase relationships of the system acoustics as well. Since the magnitude of the acoustic impedance for this supply system does not vary dramatically up to the Helmholtz resonance frequency, and the combustion chamber impedance is similarly flat up to the quarter-wave resonance at about 260 Hz, the variation in the Rayleigh integral may be expected to be reflected in the magnitude of the acoustic pressures.

The computed maxima for these configurations lie in the region 150–250 Hz, consistent with the observations. With propane, only the configuration with the swirler seated is predicted to excite above 300 Hz and to have a maximum at about the observed frequency of 248 Hz. At higher frequencies the positive and negative contributions from different regions of the flame tend to cancel, as seen in Fig. 10a. However, with methane in configuration 1, the greater distance between the effective origin and the peak heat release allows enough time for the Rayleigh integral (not plotted here) to become positive again in the range 300–400 Hz, which is where the second strong peak in the spectrum (at 368 Hz) was observed for these conditions.

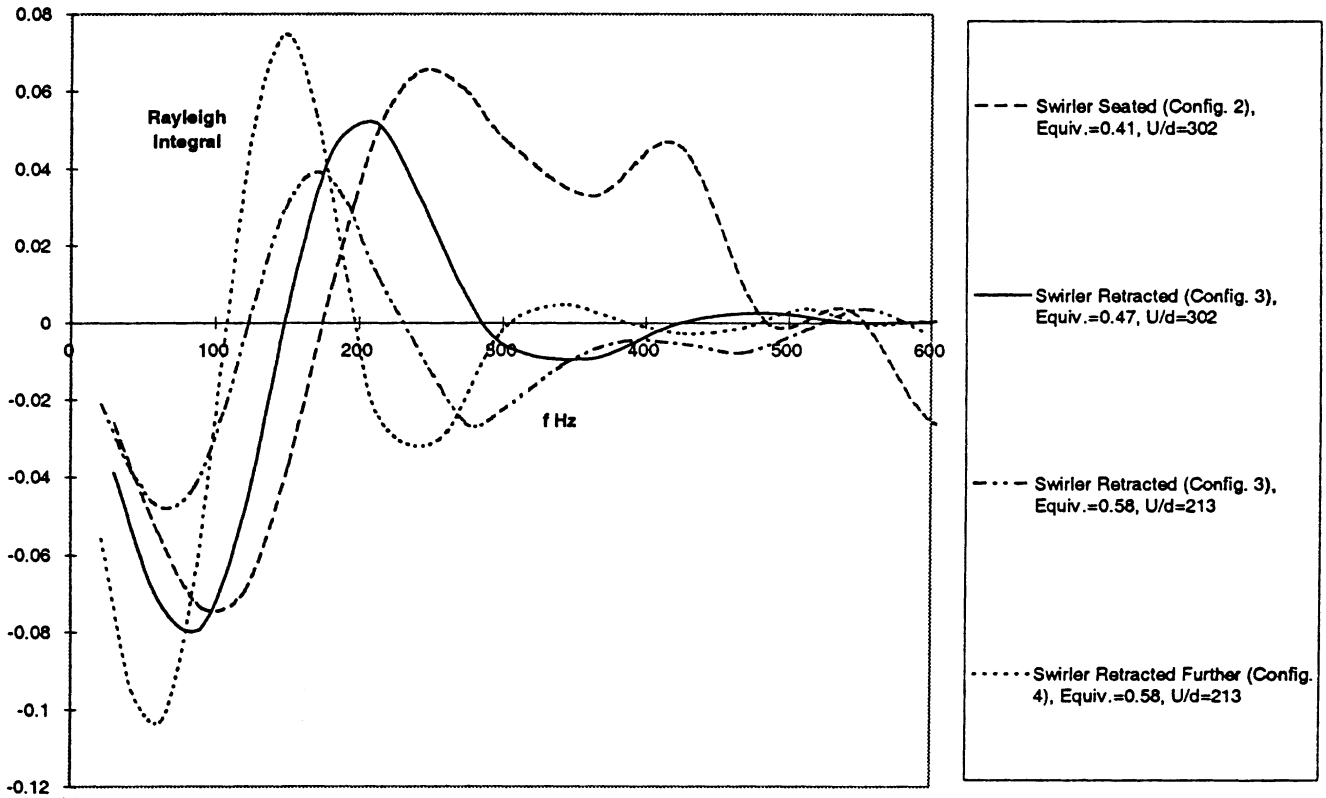
A similar model could be constructed to account for the lower-frequency mode. However, this is obviously not described by the same correlations for the magnitude and phase of the heat release as the higher-frequency mode, and so is not predicted by the Rayleigh integrals calculated for Fig. 10b.

5.5 Nature of the disturbances

While the convection-related time delay concept has been used in many thermo-acoustic calculations (see, for example, reference [**8**]) and is strongly endorsed by the present measurements, there is little real understanding of the nature of the convected disturbance. Lieuwen and Zinn [**7**] and Dowling and Hubbard [**10**] specifically



(a)



(b)

Fig. 10 (a) Axial distribution of the model contributions to the Rayleigh integral from the higher-frequency mode at $f = 272$ Hz for three burner configurations at $Q_a = 80$ l/min and one at $Q_a = 60$ l/min. (b) Values of the Rayleigh integral for the above conditions, as a function of the frequency. Excitation is possible in the higher-frequency mode at the frequencies for which the integral is positive

modelled perturbations in equivalence ratio and successfully predicted their measured instability regimes. There were certainly fluctuations in ϕ, and a possibility that fluctuations in fuel supply were important, in the present experiments. Measurements of pressure fluctuations in the fuel plenum indicated similar magnitudes and phases to those in the air outside it up to $f = 200$ Hz. This suggests little fluctuation in fuel supply, which is to be expected with the high pressure drop in the narrow pipe upstream of the plenum. However, the full magnitude of fluctuating pressure across the fuel supply trees would have given rise to a 10 per cent fluctuation in flowrate, so that some fluctuations comparable with those in air flow, which were only about 3 per cent, cannot be ruled out. Ignoring this possibility, the equivalence ratio, and hence the burning velocity in these lean conditions, will have been out-of-phase with the velocity. This is consistent with equivalence ratio fluctuations being the mechanism for the excitation of the lower-frequency modes along the axis.

Another well-known mechanism for acoustic interaction with flames is the generation of large-scale coherent vortical structures in the shear layers at the exit from the burner, as discussed by Coats [**21**]. Lin *et al.* [**22**] have studied such interactions in a lifted jet flame, and Poinsot *et al.* [**23**], McManus *et al.* [**24**], Yu *et al.* [**25**] and Schadow and Gutmark [**26**] and others, the interactions in shear layers. It is hypothesized that this is the mechanism for the higher-frequency mode of excitation, the coherent structures generated in the burner when the acoustic velocity is at the peak of its cycle giving rise to an enhancement of the heat release downstream which is in phase with their passage. This speculation can only be confirmed by visualizing or measuring the details of the flow structure, and this is the next phase of the investigation.

6 CONCLUSIONS

1. The flame in this particular swirl burner was susceptible to two distinct modes of self-excitation. These were characterized by different distributions of fluctuating heat release and different relationships between its phase and the acoustic velocity fluctuations at the burner exit.
2. The modes were distinguished by their Strouhal numbers, since the frequencies scaled approximately on the mean burner exit velocity, and so they could be loosely designated 'lower-frequency' and 'higher-frequency' modes.
3. The response of different burner configurations in the higher-frequency mode were analysed in terms of four parameters, which characterize the position and spread of the axial distribution of heat release fluctuations, and the origin and effective velocity of convection of disturbances.
4. The combination of these parameters with the acoustic impedance of the supply system in the Rayleigh integral was used to determine the frequency bands within which excitation was possible, and these were consistent with the data.
5. Depending on the precise combination of thermo-acoustic parameters, it may be desirable either to lengthen or to shorten the axial heat release distribution, so as to change the convective time delays. Retracting the swirler to produce an annular sheath of unswirled premixture, reducing the equivalence ratio, and changing from propane to methane all lengthened the distribution.
6. It is possible that the lower-frequency mode was due to fluctuations in the equivalence ratio associated with the acoustic fluctuations in the air supply, and that the higher-frequency mode was due to the periodic generation of large scale vortical structures in the shear layer. Further measurements are required to confirm this.

ACKNOWLEDGEMENTS

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